

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460658.648 +/- 0.01 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.045
Transit depth (Rp/Rs)^2	1.96 +/- 1.25 [%]
Orbital Inclination (inc)	86.45 +/- 2.05
Airmass coefficient 1 (a1)	0.935 +/- 0.01
Airmass coefficient 2 (a2)	0.045 +/- 0.039
Scatter in the residuals of the lightcurve fit is	1.08 %
Best Comparison Star	#2 - [398, 335]
Optimal Aperture	3.41
Optimal Annulus	10.21
Transit Duration (day)	0.06 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.14 $\pm$ 0.045 vs 0.11 (deviation 0.67 σ)
Duration fit vs published	0.06 d vs 0.1208 d (deviation 3.2 σ)
Mid-time O-C	-25.5 min
Depth SNR	3.1
Residual scatter	1.08 %

Light curve

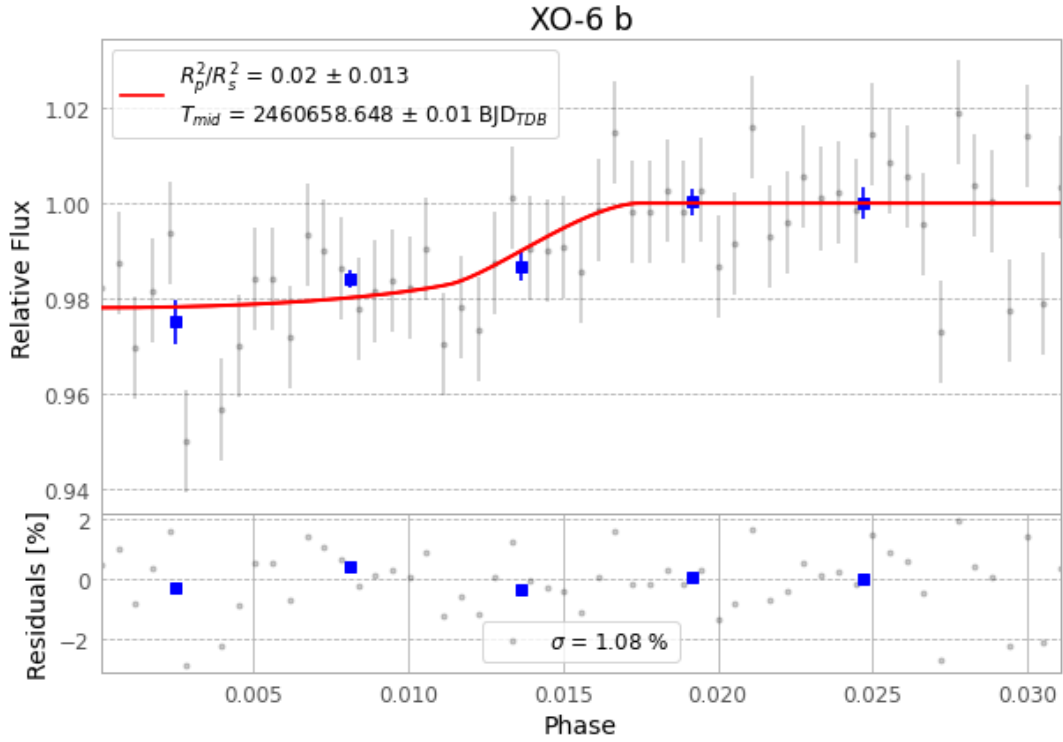


Figure 1 — FinalLightCurve_XO-6 b_2024-12-13.png (SHA-256 8d0907f4a5af3959...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [448, 184], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 243e05e2e6921270...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

57 FITS frames (37 MB), XO-6241214032715.FITS → XO-6241214061516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-6-241214.log (1302f85df77e...), FinalLightCurve_XO-6 b_2024-12-13.png (8d0907f4a5af...), FinalLightCurve_XO-6 b_2024-12-13.csv (5fee6f3eddf...)

Compute resources

Wall time 92.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 23:52Z.